Supplementary Material: A Bedside Decision Support System for Disorders of Consciousness: Multi-Scale Temporal Dynamics and Multicenter Validation

Hongjie Chen[1], Jianming Chen[2], Guanyu Xiong[2], Fang Li[2], and Fang Duan[1,*]

SUPPLEMENTARY CLINICAL AND METHODOLOGICAL NOTES

A. CRRC Cohort Recruitment and Ethics Compliance

The prospective inpatient hospital cohort was evaluated at the China Rehabilitation Research Center (CRRC / Beijing Bo'ai Hospital, Capital Medical University Rehabilitation College). Patients presenting with disorders of consciousness secondary to severe acute brain injury (11 intracerebral hemorrhage [ICH], 4 traumatic brain injury [TBI]) undergoing inpatient neurorehabilitation were enrolled under institutional review board ethical approval (Medical Ethics Committee of China Rehabilitation Research Center, Approval No. 2025-118-01; cohort characteristics and auditory paradigm screening published in Duan et al. [1]). Written informed surrogate consent was obtained from legally authorized representatives prior to telemetry recording in accordance with the Declaration of Helsinki. Diagnostic categorization into minimally conscious state (MCS, $N = 6$) and vegetative state / unresponsive wakefulness syndrome (VS/UWS, $N = 9$) was confirmed via serial Coma Recovery Scale-Revised (CRS-R) evaluations administered by certified neurorehabilitation clinicians (CRS-R scores ranging from 2 to 23). To preclude acute pharmacological confounding, all telemetry recordings adhered to a strict pre-recording clinical protocol requiring a ≥ 48 h sedation-free washout window, ensuring that observed electrophysiological dynamics reflected intrinsic thalamocortical mesocircuit state rather than transient sedative suppression.

B. Telemetry Acquisition and Signal Preprocessing

Bedside electroencephalography was acquired using a research-grade NeuSen NSM2 amplifier / EEGCap32-pony system configured at a native sampling rate of 1,000 Hz with active Ag/AgCl electrodes positioned according to the international 10–20 montage (19 channels; see Duan et al. [1]). Continuous recordings spanned intermediate monitoring intervals (mean 19.53 ± 7.96 minutes; range 7.27 to 39.07 minutes). Raw continuous signals were re-referenced to the Common Average Reference (CAR) and segmented into non-overlapping 4-second blocks (4,000 samples per block).

To ensure strict zero-leakage evaluation, artifact screening was executed at the single-record level: blocks with peak-to-peak amplitude exceeding $500\,\mu$V were flagged as artifactual gaps ($\mathbb{I}_{\text{adj}} = 0$). Homologous electrodes matching the discovery montage (F3, F4, C3, C4, O1, O2) were extracted.

Two subjects (CRRC_008 and CRRC_009) exhibited persistent high-impedance drift isolated to the occipital channel O_2. To evaluate clinical resilience objectively, two complementary validation protocols were executed:

1) **Protocol 1 (Strict Clean 6-Channel Protocol,** $N = 13$**):** Excluded the two subjects with localized O_2 artifacts, evaluating all records with fully intact 6-channel arrays.
2) **Protocol 2 (Full Sensitivity Protocol,** $N = 15$**):** Retained all 15 enrolled subjects under an intent-to-screen sensitivity protocol regardless of single-channel electrode impedance.

C. Matched Cross-Center Preprocessing and Operating Points

All matched comparisons use the same feature matrix and source training records for every model. The primary matrix is $[c_{\text{global}}, -\text{CV}_\theta, \rho_1, r_{\text{residual}}]$. Source-only inner three-fold cross-validation selects ridge and logistic regularization; LDA uses analytical shrinkage. Full-source feature signs are $\mathbf{s}_{\text{source}} = [+1, +1, +1, -1]^\top$. Model fitting uses source-standardized, sign-coded coordinates.

Frozen-source transfer reuses source means, scales, and signs. Target-standardized transfer computes means and scales from the unlabeled target batch while retaining source signs. CRRC target-aligned transfer uses the same target-standardized matrix and $\mathbf{s}_{\text{target}} = [-1, +1, +1, +1]^\top$ for all models. In raw standardized coordinates, this corresponds to $\text{diag}(-1, +1, +1, -1)$ applied consistently to the target inputs of source-trained supervised models. Target labels enter only the final evaluation. The direction rule is specified for the subacute clinical setting; its prospective deployment requires the rule and calibration batch to be fixed before evaluating later patients.

The primary four-feature score is the arithmetic mean. The fixed-truncation variant averages $\text{clip}(s_k z_k, -1.5, 1.5)$ and uses the same calibration state. It yields 0.9291 AUROC in discovery and 0.7963 on the full CRRC cohort, versus 0.9324 and 0.7778 for the untruncated score. Supplementary Table S2 separates this scoring change from the transfer protocols.

[1]Hongjie Chen and Fang Duan are with the College of Information Science and Engineering, Huaqiao University, Xiamen 361021, China.

[2]Jianming Chen, Guanyu Xiong, and Fang Li are with the China Rehabilitation Research Center, Beijing Bo'ai Hospital, Capital Medical University, Beijing, China.

*Corresponding author: Fang Duan (e-mail: nkfetsh@gmail.com).

TABLE S1

DE-IDENTIFIED INDIVIDUAL COHORT CHARACTERISTICS, FEATURE PROFILES, AND CROSS-CENTER BENCHMARK PERFORMANCE FOR THE CHINA REHABILITATION RESEARCH CENTER (CRRC / BEIJING BO'AI HOSPITAL) INPATIENT COHORT ($N = 15$; 15 MALES; 6 MCS [INCLUDING 1 EMCS], 9 VS/UWS; 11 ICH, 4 TBI; MEAN AGE 51.7 ± 12.3 YEARS). CLEAN 6-CHANNEL RECORDS ($N = 13$; 6 MCS [INCLUDING 1 EMCS], 7 VS/UWS) EXCLUDE CRRC_008 AND CRRC_009 DUE TO CONTINUOUS MECHANICAL ELECTRODE DRIFT ON O_2.

Subject ID	Diagnosis	CRS-R	Lesion Laterality	Duration (s)	Retention	c_{global}	$-\text{CV}_\theta$	ρ_1	r_{residual}
CRRC_001	MCS	11	Bilateral intact	605.0	98.7%	0.1834	-1.3927	+0.1118	+0.5878
CRRC_002*	MCS^{+}* (EMCS)	23	Right hemisphere	1236.0	98.7%	0.5343	-0.3331	+0.3201	+0.2064
CRRC_003	MCS	10	Left hemisphere	1225.0	97.1%	0.8409	-0.3682	+0.1451	+0.3646
CRRC_004	MCS	7	Left hemisphere	1221.0	100.0%	0.5993	-0.3748	-0.0378	+0.1391
CRRC_005	MCS	17	Bilateral intact	845.0	100.0%	0.4906	-0.6205	+0.3850	+0.3328
CRRC_006	MCS	9	Right hemisphere	1216.0	97.4%	0.5173	-0.4568	+0.5073	+0.2188
CRRC_007	VS/UWS	7	Right hemisphere	1896.0	99.8%	0.8119	-0.4731	+0.0105	+0.3776
CRRC_008†	VS/UWS	2	Bilateral intact	1213.0	100.0%	0.7253	-0.6384	+0.0337	+0.3048
CRRC_009†	VS/UWS	3	Bilateral intact	1209.0	99.7%	0.6213	-0.6091	+0.0614	+0.1660
CRRC_010‡	VS/UWS	2	Bilateral intact	637.0	21.4%	0.7294	-0.4525	+0.2401	-0.0657
CRRC_011	VS/UWS	7	Bilateral intact	1237.0	99.0%	0.4154	-0.4882	+0.4003	+0.3513
CRRC_012	VS/UWS	7	Bilateral intact	436.0	99.1%	0.5607	-0.3266	+0.0680	+0.0339
CRRC_013	VS/UWS	7	Left hemisphere	2344.0	99.5%	0.5517	-0.4399	+0.1056	+0.2702
CRRC_014	VS/UWS	7	Right hemisphere	1215.0	86.8%	0.7556	-0.6032	+0.6941	+0.1504
CRRC_015	VS/UWS	4	Left hemisphere	1040.0	100.0%	0.6218	-0.7362	-0.0033	+0.3001

Cross-Center Performance Comparison Across Evaluation Protocols and Model Architectures:

• **Protocol 1 (Clean 6-Ch, $N = 13$; 6 MCS/EMCS, 7 VS/UWS)**: The two-feature static reference yields 0.5952 AUROC. With shared target standardization and direction alignment, four-feature linear scoring, fixed truncation, and source-tuned ridge yield 0.7143; source-tuned logistic regression and LDA yield 0.6905. Frozen-source linear transfer yields 0.5238.

• **Protocol 2 (Full Cohort, $N = 15$; 6 MCS/EMCS, 9 VS/UWS)**: The two-feature static reference yields 0.6481 AUROC. Under matched target alignment, four-feature linear scoring yields 0.7778 and fixed $c = 1.5$ truncation yields 0.7963 (subject-level Bayesian 95% interval $[0.5213, 0.9630]$). Source-tuned ridge, logistic regression, and LDA yield 0.7407, 0.7593, and 0.7593. All-model protocol comparisons appear in Table S2; operating thresholds are defined in Note C.

*Patient CRRC_002 presented with unilateral right hemisphere stroke and severe left motor deficit; scored 23 on the CRS-R with functional communication intact, meeting standard diagnostic criteria for Emergence from MCS (EMCS), and was retained in the conscious/emerging class (MCS^{+}) for binary triage screening.

†CRRC_008 and CRRC_009 exhibit documented O_2 electrode artifacts; excluded in Protocol 1 ($N = 13$) and retained in Protocol 2 ($N = 15$).

‡CRRC_010 exhibited low retention (21.4%, 34 clean blocks) due to frequent head motion; retained under Protocol 2 intent-to-treat screening.

All records acquired via NeuSen NSM2 / EEGCap32-pony (Neuracle, 1000 Hz, 4-s epochs [4,000 samples], BioSemi-compatible channel layout; Ethics Approval No. 2025-118-01; cohort details reported in Duan et al. [1]) and evaluated across homologous channels (F3, F4, C3, C4, O1, O2). De-identified tabular feature matrices are open-source.

TABLE S2

MATCHED-INPUT MODEL COMPARISONS (AUROC). ALL METHODS SHARE FEATURE COLUMNS, SOURCE-ONLY PARAMETER SELECTION, SCALING, DIRECTION CODING, AND SOURCE-TRAINING MEDIAN THRESHOLDS WITHIN EACH PROTOCOL. TRUNCATION IS A FIXED SCORING-RULE VARIANT ($c = 1.5$). TARGET STANDARDIZATION AND DIRECTION ALIGNMENT ARE UNLABELED BATCH ADAPTATIONS.

Cohort and protocol	Linear	Clipped	Ridge	Logistic	LDA
Mendeley, 25 outer folds	0.9324	0.9291	0.8813	0.8964	0.8769
CRRC 15, source frozen	0.5741	0.5741	0.5556	0.5556	0.5556
CRRC 15, target standardized	0.5000	0.5000	0.5185	0.5185	0.5370
CRRC 15, target aligned	0.7778	0.7963	0.7407	0.7593	0.7593
CRRC 13, target aligned	0.7143	0.7143	0.7143	0.6905	0.6905
Figshare 50, D_1, source frozen	0.7691	0.7606	0.7759	0.7674	0.7742
Figshare 50, D_1, target standardized	0.7657	0.7606	0.7708	0.7589	0.7691

Primary decisions use each model's source-training score median. For full-cohort CRRC truncated scoring, sensitivity, specificity, and balanced accuracy are 0.6667, 0.7778, and 0.7222. An auxiliary unlabeled target-median threshold gives 0.8333, 0.6667, and 0.7500. For the Figshare jump-rate score, the source threshold gives 1.0000, 0.1290, and 0.5645, while the auxiliary target median gives 0.8421, 0.7097, and 0.7759. These policies have separate archived thresholds; neither is selected by scanning target labels.

The repository provides per-subject scores, primary decisions, auxiliary operating points, all source and target transforms, regularization-selection traces, and input hashes in results/matched_model_comparison_v1/. Paired uncertainty uses 10,000 subject-level Dirichlet draws, sharing weights across methods and repeated folds and conditioning on the fitted predictions.

REFERENCES

[1] F. Duan *et al.*, "Auditory paradigm-dependent EEG feature screening and representation for distinguishing consciousness states in male patients with disorders of consciousness," *Frontiers in Neurology*, vol. 17, p. 1927313, 2026.